

ULLEUNGDO, Korea, Republic of

WMO: 471150

Lat: 37.4814N

Lon: 130.8986E

Elev: 735

StdP: 14.31

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	25.3	9.3	9.1	29.1	12.7	10.8	30.1	26.6	32.9	23.4	34.5	8.4	20	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	85.5	76.0	83.4	75.2	81.3	73.9	77.8	82.9	76.5	81.3	75.3	79.8	6.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.4	141.5	81.2	75.1	135.4	79.8	73.9	129.7	78.6	41.9	82.9	40.7	81.5	39.5	80.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.1	21.5	18.8	DB	18.2	89.3	3.6	3.1	15.6	91.6	13.5	93.3	11.5	95.1	8.9	97.3	(4)
			WB	16.5	79.8	3.3	1.8	14.2	81.1	12.2	82.2	10.4	83.2	8.0	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	35.0	36.6	42.8	52.4	60.8	66.7	73.4	75.8	68.5	60.5	50.1	39.8	(6)
		DBStd	14.96	5.65	6.22	6.53	6.38	5.48	4.62	5.12	4.68	4.11	4.94	6.71	6.62	(7)
		HDD50	1529	465	375	236	46	1	0	0	0	3	82	321		(8)
		HDD65	4372	930	794	687	381	152	34	3	0	150	447	782		(9)
		CDD50	3466	0	1	14	117	337	500	724	801	556	328	85	3	(10)
		CDD65	835	0	0	0	2	22	84	262	337	118	10	0	0	(11)
		CDH74	4062	0	0	0	1	55	181	1433	2149	241	2	0	0	(12)
		CDH80	723	0	0	0	0	6	13	255	433	16	0	0	0	(13)
(14)	Wind	WSAvg	8.1	7.8	8.2	9.2	9.6	8.7	7.0	8.1	7.3	7.3	7.8	8.2	7.8	(14)
(15)	Precipitation	PrecAvg	53.9	4.9	3.4	2.9	3.5	3.8	4.2	5.8	6.2	6.1	3.7	4.4	4.6	(15)
(16)		PrecMax	88.1	13.6	9.4	7.1	15.6	12.8	12.1	20.7	24.5	18.4	10.3	16.9	12.2	(16)
(17)		PrecMin	30.3	0.7	0.9	0.3	0.7	0.8	0.7	0.8	0.9	0.5	0.2	0.9	1.0	(17)
(18)		PrecStd	13.2	2.5	1.6	1.5	2.4	2.4	2.5	3.7	4.5	3.9	2.5	2.7	2.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	51.0	55.2	62.9	72.2	79.6	81.6	87.4	89.5	82.2	73.0	66.9	57.5		(19)
(20)		MCWB	46.0	50.2	51.1	57.1	60.3	68.6	75.9	77.9	75.4	65.4	59.2	52.2		(20)
(21)		DB	47.3	51.2	58.1	67.7	74.9	78.2	85.1	86.4	78.9	70.7	63.8	53.6		(21)
(22)		MCWB	41.9	45.7	48.9	54.7	59.9	67.5	75.2	77.1	72.5	64.0	57.3	48.3		(22)
(23)		DB	45.1	48.1	54.9	64.6	72.1	75.9	82.9	84.3	76.6	68.8	61.4	51.3		(23)
(24)		MCWB	39.9	43.0	47.2	54.1	59.7	66.8	74.5	76.2	70.7	62.6	55.4	45.8		(24)
(25)		DB	42.9	45.4	52.4	61.9	69.5	73.8	80.9	82.4	74.8	67.0	59.2	48.9		(25)
(26)		MCWB	37.8	40.5	46.3	52.8	58.6	65.9	73.6	75.2	69.5	61.1	53.4	43.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	47.7	52.1	54.0	62.1	67.5	73.1	78.8	80.6	76.8	68.9	62.2	53.0		(27)
(28)		MCDB	49.8	54.3	59.0	67.0	73.6	77.7	84.1	86.4	80.8	71.2	64.7	55.6		(28)
(29)		WB	43.0	47.1	51.5	58.4	64.3	70.7	76.9	78.6	74.3	66.0	58.7	49.3		(29)
(30)		MCDB	46.7	50.0	55.1	64.2	70.1	74.9	82.7	83.7	77.4	69.1	62.6	52.8		(30)
(31)		WB	40.4	43.4	49.2	56.2	62.4	69.1	75.5	77.4	72.3	63.8	56.3	46.4		(31)
(32)		MCDB	44.3	47.3	53.8	61.7	68.5	73.1	80.9	82.1	75.3	67.5	60.5	50.9		(32)
(33)		WB	38.3	40.7	46.8	54.1	60.7	67.7	74.4	76.3	70.5	62.1	53.9	43.8		(33)
(34)		MCDB	42.2	45.2	51.5	59.4	66.3	71.7	79.5	80.8	73.7	66.0	58.6	48.3		(34)
(35)	Mean Daily Temperature Range	MDBR	7.2	8.3	10.1	11.2	10.8	9.0	7.4	7.8	8.1	8.6	8.6	7.7		(35)
(36)		5% DB	9.9	11.8	13.2	14.4	13.8	11.9	9.8	10.1	10.1	10.1	10.5	10.1		(36)
(37)		MCWBR	8.6	10.1	10.6	9.8	9.8	8.2	6.5	6.8	8.0	8.3	8.8	8.9		(37)
(38)		5% WB	9.6	11.5	12.3	12.3	12.2	10.0	9.1	9.2	8.7	9.1	10.3	10.1		(38)
(39)		MCWBR	9.1	10.4	11.1	10.0	10.3	7.7	6.7	6.6	7.7	8.3	9.6	9.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.409	0.447	0.514	0.539	0.542	0.560	0.551	0.502	0.441	0.435	0.418	0.405		(40)
(41)		taud	2.059	1.985	1.841	1.835	1.891	1.927	2.027	2.178	2.285	2.151	2.118	2.074		(41)
(42)		Ebn at Noon	233	240	236	239	242	237	238	247	257	242	229	223		(42)
(43)		Edh at Noon	39	48	61	65	62	60	54	46	39	40	37	36		(43)
(44)	All-Sky Solar Radiation	RadAvg	576	861	1282	1621	1762	1745	1584	1595	1315	1076	712	524		(44)
(45)		RadStd	52	75	94	131	192	145	185	151	122	92	74	40		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	-2.21	N/A	N/A	N/A	N/A	N/A	+139	N/A	(47)

Nomenclature: See separate page